

Collective Habit as Infrastructure: A Computational Model of Morphic Resonance via Multi-Agent Reinforcement Learning

Technical Report — BUTTERS Project

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April 2026

Abstract

We present a computational model of Rupert Sheldrake’s morphic resonance hypothesis, implemented as a multi-agent reinforcement learning simulation of the McDougall rat maze experiment (1920–1954). The simulation places populations of tabular Q-learning agents in a procedurally generated maze across 60 generations under four experimental conditions: no inheritance (control), genetic inheritance only, morphic field only, and both mechanisms combined. The morphic field is modeled as a performance-weighted shared Q-table that accumulates knowledge from all past agents and emits it as a scaled prior for new agents, with no genetic or physical channel of transmission. A fifth condition — the *displaced control* — drops fresh agents with zero ancestry into a mature morphic field at generation 60. Results show: (1) the control condition remains flat at 400 steps across all generations; (2) genetic-only converges to 56 steps by generation 5 (86% improvement); (3) morphic-only improves to 245 steps (39%) with no genetic inheritance; (4) displaced control agents immediately score 297 steps versus the 400-step baseline — a 26% advantage on their first generation with zero ancestry. We argue that the simulation does not demonstrate the existence of morphic resonance in nature, but establishes three things: the hypothesis is formally coherent, the predicted data pattern is distinctive and detectable, and the displaced control constitutes the most powerful experimental design for testing it. We further show that the morphic field architecture — performance-weighted absorption, scaled emission, and blending with local learning — constitutes a practical design pattern for multi-agent knowledge transfer, cold-start mitigation, and stigmergic coordination in AI systems.

Keywords: morphic resonance, Q-learning, multi-agent systems, collective memory, displaced control, stigmergy, knowledge transfer, cold start problem

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1 Introduction

Rupert Sheldrake’s hypothesis of morphic resonance [1,2] proposes that natural systems inherit a collective memory from all previous systems of the same kind, transmitted not through known physical mechanisms but through *morphic fields* — non-material regions of influence that shape form, development, and behavior. The hypothesis has been a lightning rod for controversy since its publication: *Nature* editor John Maddox called Sheldrake’s first book “the best candidate for burning” [4], while supporters argue that orthodox biology lacks an account of how identical genomes produce radically different cell types, or why the so-called “laws” of nature should be eternal rather than habitual [3].

The hypothesis remains untested in a decisive way, in part because it is difficult to formalize. What *exactly* is a morphic field? How does similarity-based resonance operate? What data pattern would we expect to see if it were real, and how would we distinguish that pattern from genetic selection, environmental drift, or statistical artifact?

This paper takes a computational approach. We introduce a *value-function-level sharing mechanism* for multi-agent RL that sits between experience-sharing methods (SUPER [21]) and weight-sharing methods (FedAvg [15]). We call this the *morphic field* architecture — inspired by, though not endorsing, Sheldrake’s morphic resonance hypothesis. Specifically, we ask:

1. Can a “morphic field,” formalized as a shared, performance-weighted Q-table with no genetic channel, transmit useful learning across agent generations?
2. What is the distinctive data pattern that would result — the *signature* an experimentalist should look for?
3. Does the displaced control test (fresh, unrelated agents dropped into a mature field) constitute a viable experimental design?
4. Does the resulting architecture have practical value for multi-agent AI systems, independent of Sheldrake’s metaphysics?

Our simulation is built on the McDougall rat maze experiment (1920–1954) [5], chosen because it is the most quantitative, generational, and independently replicated of the six major evidence categories for morphic resonance (Section 2.2). We model rats as tabular Q-learning agents, generations as population cohorts, genetic inheritance as Q-table crossover with mutation, and the morphic field as an accumulated, performance-weighted Q-table that emits a scaled prior to unrelated agents.

Contributions.

1. A formal computational model of morphic resonance using multi-agent reinforcement learning (Section 3).
2. Empirical results across five experimental conditions demonstrating the displaced control signature (Section 4).
3. An epistemological analysis of what the simulation does and does not prove (Section 5).
4. Identification of the morphic field as a practical design pattern for multi-agent knowledge transfer, cold-start mitigation, and stigmergic coordination (Section 6).

Paper organization. Section 2 reviews Sheldrake’s hypothesis, the McDougall experiment, and related work in multi-agent learning. Section 3 describes the simulation architecture. Section 4 presents experimental results. Section 5 analyzes what the results mean. Section 6

connects the morphic field to practical AI design patterns. Section 7 discusses limitations. Section 8 concludes.

2 Background and Related Work

2.1 Morphic Resonance

A *morphic field* (from Greek *morphe*, form) is, in Sheldrake’s framework, a field of information that shapes the form, development, and behavior of a system [1]. Morphic fields are localized within and around *morphic units* — any unit of organization, from subatomic particles to organisms to social groups — and are organized in a nested hierarchy (a *holarchy*, per Koestler [7]).

Morphic resonance is the process by which information is transmitted from past morphic units to present ones on the basis of *similarity*: the more similar a present system is to past systems, the stronger the resonance. The mechanism is explicitly non-energetic — no photons, no spacetime curvature, no known carrier — and non-local in both space and time. Sheldrake frames this as a radical departure from the assumption that the “laws” of nature are fixed and eternal, proposing instead that they are *habits* that accumulate through repetition [2].

Key terminology. **Morphic unit:** any system with its own morphic field. **Morphogenetic field:** the subset governing embryonic development (the original concept from Waddington and others, which Sheldrake generalizes). **Behavioral field:** governs learned or instinctive behavior. **Chreode:** a canalized pathway of development (Waddington’s term, adopted by Sheldrake for habitual trajectories in morphic fields).

How morphic fields differ from known physical fields. (A) No known energy is transmitted. (B) No inverse-square attenuation with distance. (C) Influence is non-local in time: past systems influence present ones without propagation delay. (D) Specificity is by similarity, not proximity. These properties make the hypothesis difficult to test but also generate distinctive predictions.

2.2 The McDougall Rat Maze Experiment

William McDougall conducted maze-learning experiments with Wistar rats at Harvard from 1920 to 1954 [5]. Rats chose between a lighted gangway (electric shock) and an unlighted gangway (safe). The first generation averaged approximately 165 errors; by the 30th generation, this had fallen to approximately 20 errors. McDougall interpreted the improvement as Lamarckian inheritance — the inheritance of acquired characteristics.

The Agar–Crew replication. Agar, Crew, and Mitchell at Edinburgh ran a parallel study from 1934 to 1954, maintaining both a trained line and, crucially, a *control line* never selected for maze ability [6]. The trained line improved, consistent with McDougall. But the control line *also improved at roughly the same rate*. The original researchers attributed this to environmental and methodological artifacts accumulated over 20 years.

Sheldrake’s reinterpretation. Sheldrake argues that the control-line improvement is the key datum: if rats never selected for maze ability also improve, something other than genetics is transmitting the learning [2]. This is the *morphic resonance signature* — the prediction that our simulation is designed to test.

Conventional counter-explanations. Laboratory condition changes over two decades, genetic drift, unconscious handling changes, and statistical fluctuation have all been proposed. No consensus exists on which explanation is correct.

2.3 Simulatability Ranking

Of the six major evidence categories for morphic resonance — rat maze learning, crystal formation, blue tit milk-bottle opening, staring detection, dog telepathy, and telephone telepathy — we selected the McDougall rat maze because it is:

1. **Quantitative and generational:** a clear measurable (errors per trial) across successive generations, mapping directly to agents and iterations in code.
2. **Has the control-line prediction:** the most striking claim is not that trained rats’ offspring improved, but that *unrelated* control lines also improved.
3. **Independently replicated:** Agar and Crew ran follow-up studies and reportedly found similar trends, though interpretations remain disputed.

The remaining evidence categories are weaker for simulation: crystal formation requires modeling poorly defined physics; blue tit behavior spreading is adequately explained by network diffusion; telephone telepathy reduces to a biased coin flip; staring detection is binary guessing susceptible to response bias [8]; and dog telepathy is single-subject observational.

2.4 Related Work in Multi-Agent Knowledge Transfer

The idea that agents can benefit from the experience of other agents is well established in AI, though it is not typically framed in terms of morphic resonance:

Cultural accumulation in RL. The closest conceptual work is Cook et al.’s cultural accumulation framework [20], which demonstrates generational knowledge transfer via PPO with a behavioral cloning auxiliary loss. Each generation trains on demonstrations from the previous generation, producing a “cultural ratchet.” Our approach differs by sharing distilled *value functions* rather than raw behavioral demonstrations, and operates bidirectionally across the entire population rather than unidirectionally from parent to child.

Experience sharing. SUPER [21] (NeurIPS 2023) shares full experience tuples (s, a, r, s') between agents in cooperative MARL. ERL/CERL [22] inject gradient-free evolutionary actors into a shared replay buffer alongside RL actors. The morphic field shares at a higher level of abstraction: entire Q-tables (value functions) rather than individual transitions, producing more compact transfer.

Federated RL. FEDORA [23] applies federated learning directly to RL, aggregating policy parameters across distributed agents. FedAvg [15] averages model weights uniformly. The morphic field differs in two key ways: (1) contributions are weighted by *performance*, not uniformly averaged; (2) temporal decay (Section 3.3) allows the field to forget stale knowledge, a mechanism absent from all existing FedRL methods.

Experience replay and distillation. Prioritized experience replay [12] allows a single agent to re-learn from weighted past transitions. Knowledge distillation [13] compresses the knowledge of a large model (or ensemble) into a smaller one. Our morphic field can be viewed as a population-level experience replay buffer, where contributions are weighted by performance and “distilled” into a shared initialization.

Population-based training. Population-based training (PBT) [14] evolves hyperparameters and model weights across a population of agents. Our genetic condition is a simplified PBT variant; the morphic field adds a non-genetic channel that PBT does not provide.

Positioning in the knowledge-transfer spectrum. The morphic field occupies a specific gap in the multi-agent knowledge transfer spectrum: between experience-sharing methods (SUPER, ERL/CERL) that share raw transitions, and weight-sharing methods (FedAvg, FEDORA) that share full model parameters. The morphic field shares *value functions* — intermediate-level representations that encode reward structure without requiring the receiver to re-derive value from raw experience.

Stigmergy. Stigmergy — indirect coordination through a shared environment [16] — is the closest analogue. Ant colony optimization [17] deposits pheromone on successful paths; our morphic field deposits weighted Q-values. The GraphPalace system [18] uses five-type pheromone stigmergy for memory retrieval, operating on the same principle in embedding space rather than Q-table space.

2.5 Bioelectric Morphogenetic Fields

Michael Levin’s laboratory at Tufts University has demonstrated that *real* morphogenetic fields exist at the cellular level — bioelectric signal patterns that guide development, regeneration, and collective cell behavior [24, 25].

Manicka & Levin 2025. In a computational study using Active Inference [26] — the same framework used in GraphPalace’s agent architecture — Manicka and Levin [25] show how bioelectric networks create fields that cells “read” to make developmental decisions. The mechanism is structurally identical to our morphic field: cells emit bioelectric signals → signals accumulate in the tissue → new cells read the accumulated field to decide their fate. This is absorb/emit/blend at the cellular level.

Levin 2025. In “The Multiscale Wisdom of the Body” [24], Levin argues that biological intelligence operates at multiple scales: molecular → cellular → tissue → organism, with dynamic rewritable memory at each level. The morphic field in our simulation operates at the same abstraction: individual agent → population → field → new agent.

Significance. Levin’s work provides a mainstream scientific bridge between Sheldrake’s controversial hypothesis and our engineering implementation. Bioelectric morphogenetic fields are published in *Nature*, *Cell*, and *BioEssays* — they are not fringe science. The morphic field in our simulation can be understood as a computational analogue of bioelectric patterning, independent of whether one endorses Sheldrake’s metaphysics.

2.6 Philosophical Context

Sheldrake’s hypothesis sits within a broader tradition of process philosophy and holistic biology.

Whitehead and objective immortality. Alfred North Whitehead’s process philosophy [9] provides the most precise philosophical mapping to our morphic field architecture. Three key concepts correspond to specific computational operations:

1. **Objective immortality → absorption.** When a Whiteheadian “actual occasion” perishes, it achieves *objective immortality* — it becomes permanent data available for future

prehension. In the morphic field, when an agent completes training and is discarded, its Q-table is absorbed into the field. The agent “perishes” but its knowledge persists.

2. **Prehension** → **emission**. Whitehead’s “prehension” is how an actual occasion grasps past entities. Positive prehension (inclusion) maps to `field.emit()` — the new agent grasps accumulated knowledge. Negative prehension (exclusion) maps to performance weighting — low-performing contributions are effectively excluded.
3. **Concrescence** → **blending**. The process by which multiple prehensions integrate into a unified experience maps to the blend operation: $\mathbf{Q}_{\text{init}} = 0.5 \cdot \mathbf{Q}_{\text{genetic}} + 0.5 \cdot \mathbf{Q}_{\text{morphic}}$.

This is not a loose analogy but a structural isomorphism. Whitehead’s metaphysics and our computational architecture share the same data flow: entities arise, contribute to a shared medium, perish, and inform future entities through that medium.

Bohm. David Bohm’s implicate order [10] posits a deeper enfolded reality beneath the manifest (explicate) world. Morphic fields could be understood as structures within the implicate order, with resonance as the process by which enfolded information unfolds into new systems.

Jung. Carl Jung’s collective unconscious [11] — an inherited stratum of shared archetypes beneath individual consciousness — is a psychological parallel. Morphic resonance generalizes the concept from human psyche to all of nature.

These frameworks are invoked here not as evidence for morphic resonance, but as context for the kind of claim being modeled: that collective memory is a fundamental property of nature, not a special property of brains.

3 Simulation Architecture

The simulation (`morphic_sim.py`, 493 lines of Python) models the McDougall experiment as a multi-agent reinforcement learning system with five experimental conditions. Figure 1 shows the overall data flow.

3.1 Maze Environment

The maze is generated via recursive backtracker (depth-first search) with `seed=42`. Cell dimensions are 7×7 , yielding a 15×15 grid (cells plus walls). The start position is (1, 1) and the goal is (13, 13).

$$\text{grid}[r, c] = \begin{cases} 1 & \text{wall} \\ 0 & \text{open passage} \end{cases} \quad (1)$$

The maze is deterministic: all experimental conditions share the same maze topology, isolating the effect of the knowledge-transfer mechanism.

3.2 Q-Learning Agent

Each agent is a tabular Q-learning agent operating on the grid:

- **State space:** (r, c) positions on the 15×15 grid.
- **Action space:** 4 cardinal directions (up, down, left, right).
- **Q-table:** $15 \times 15 \times 4$ array of float64 values.

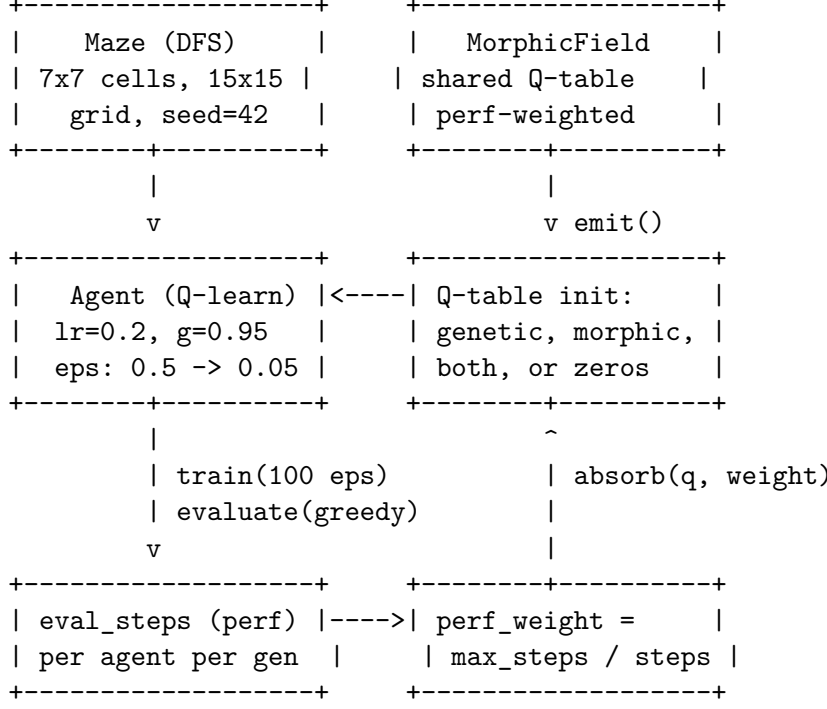


Figure 1: Simulation data flow. Each generation: (1) agents are initialized from genetic parents, morphic field, both, or zeros; (2) agents train for 100 episodes; (3) agents are evaluated greedily; (4) Q-tables are absorbed into the morphic field weighted by performance; (5) elite parents are selected for the next generation’s genetic channel.

- **Hyperparameters:** learning rate $\alpha = 0.2$, discount factor $\gamma = 0.95$, ϵ decays linearly from 0.5 to 0.05 over training episodes.
- **Rewards:** +100 for reaching goal, -1 per step, -5 for wall collisions.
- **Training:** 100 episodes per agent, max 400 steps per episode.
- **Evaluation:** greedy ($\epsilon = 0$), returns steps to goal or 400 (max) if unsolved.

The Q-learning update rule is standard:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (2)$$

3.3 Morphic Field

The morphic field is the core contribution. It is modeled as a `MorphicField` class holding an accumulated Q-table of the same shape as agent Q-tables ($15 \times 15 \times 4$).

Definition 1 (Morphic Field). *A morphic field $\mathcal{F}$ is a tuple $(\mathbf{T}, W, n)$ where $\mathbf{T} \in \mathbb{R}^{H \times W \times A}$ is the accumulated Q-table, $W \in \mathbb{R}^+$ is the total weight, and $n \in \mathbb{N}$ is the contributor count.*

Absorption. When an agent completes training and evaluation, its Q-table is absorbed into the field, weighted by performance:

$$\mathbf{T} \leftarrow \mathbf{T} + \mathbf{Q}_i \cdot w_i, \quad W \leftarrow W + w_i, \quad n \leftarrow n + 1 \quad (3)$$

where the performance weight is:

$$w_i = \begin{cases} \frac{\text{max_steps}}{\text{eval_steps}_i} & \text{if agent solved the maze} \\ 0.01 & \text{otherwise} \end{cases} \quad (4)$$

This ensures that agents who solve the maze quickly contribute proportionally more, while agents who fail contribute near-zero weight. The field does not fill with noise from failed agents.

Emission. When a new agent is initialized from the field, it receives a scaled copy of the weighted average:

$$\mathbf{Q}_{\text{init}} = \frac{\mathbf{T}}{W} \cdot s \quad (5)$$

where $s = 0.5$ is the emission strength. This gives new agents a *nudge* in the direction of past solutions without fully determining their behavior — they must still learn.

Temporal decay. To model pheromone evaporation in stigmergic systems, the morphic field supports exponential temporal decay. Before each absorption, the existing field is decayed:

$$\mathbf{T} \leftarrow \mathbf{T} \cdot (1 - \rho), \quad W \leftarrow W \cdot (1 - \rho) \quad (6)$$

where $\rho \in [0, 1)$ is the decay rate. With $\rho = 0$ (default), the field is purely cumulative. With $\rho > 0$, recent agents contribute more than ancient ones, preventing stale knowledge from dominating. This is the key differentiator from federated RL methods, none of which implement temporal decay. Section 4.3 evaluates the effect of different decay rates.

Key design properties. The field has the following properties that map onto Sheldrake’s description:

1. **No genetic channel:** the field operates independently of parent–child relationships. Agents absorb into and emit from the field regardless of ancestry.
2. **Similarity by task:** all agents solving the same maze “resonate” with each other through the shared Q-table shape.
3. **Cumulative:** the field strengthens with each generation, modeling Sheldrake’s claim that habits of nature deepen through repetition.
4. **Performance-weighted:** better solvers contribute more, modeling the idea that more successful forms of organization leave stronger morphic imprints.

3.4 Genetic Inheritance

The genetic channel models standard neo-Darwinian evolution:

1. **Elite selection:** agents are ranked by evaluation performance; the top 30% (minimum 2) are selected as parents.
2. **Crossover:** two random elite parents are selected; a uniform binary mask determines which Q-values come from which parent.
3. **Mutation:** Gaussian noise ($\sigma = 0.05$) is added to the child Q-table.

When both genetic and morphic channels are active, the child Q-table is a 50/50 blend:

$$\mathbf{Q}_{\text{init}} = 0.5 \cdot \mathbf{Q}_{\text{genetic}} + 0.5 \cdot \mathbf{Q}_{\text{morphic}} \quad (7)$$

3.5 Experimental Conditions

Table 1 summarizes the five experimental conditions.

Table 1: Experimental conditions. Each condition enables or disables the genetic and morphic channels independently.

Condition	Genetic	Morphic	Purpose
Control	–	–	Baseline: no inheritance
Genetic Only	✓	–	Standard evolution
Morphic Field Only	–	✓	Collective habit, no genetics
Genetic + Morphic	✓	✓	Both mechanisms
Displaced Control	–	✓*	Fresh agents in mature field

*Morphic field pre-trained for 60 generations under the morphic-only condition.

The *displaced control* is the critical test. It corresponds to Sheldrake’s prediction about the Agar–Crew control line: fresh agents with **zero** genetic connection to any prior solver are initialized using only a mature morphic field accumulated over 60 prior generations of morphic-only training. If these agents outperform the control baseline on their first generation, the field alone is transmitting learned information.

3.6 Algorithm

Algorithm 1 gives the complete procedure for running one experimental condition across generations.

4 Experimental Results

All experiments use seed 42, 60 generations (20 for displaced control), 20 agents per generation, and 100 training episodes per agent.

4.1 Main Conditions

Table 2 presents the headline results.

Table 2: Simulation results across conditions (7×7 maze, seed=42, 60 generations, 20 agents/-generation, 100 training episodes). “Steps” is the mean evaluation steps in greedy mode (lower is better; 400 = unsolved).

Condition	Gen 1 (steps)	Gen 60 (steps)	Improvement
Control	400	400	0%
Genetic Only	400	56	86%
Morphic Field Only	400	245	39%
Genetic + Morphic	400	108	73%
Displaced Control*	297	228	—

*Fresh agents, zero ancestry, mature morphic field from generation 60. Runs for 20 generations.

Algorithm 1 Run one experimental condition

Require: Maze M , condition $\in \{\text{control, genetic, morphic, both}\}$, generations G , population P , episodes E

Ensure: History $H = [h_1, \dots, h_G]$ of mean evaluation steps

```
1:  $\mathcal{F} \leftarrow \text{MorphicField}()$  if morphic enabled
2: parents  $\leftarrow \emptyset$ 
3: for  $g \leftarrow 1$  to  $G$  do
4:   for  $i \leftarrow 1$  to  $P$  do
5:      $\mathbf{Q}_g \leftarrow \emptyset, \mathbf{Q}_f \leftarrow \emptyset$ 
6:     if genetic enabled and parents  $\neq \emptyset$  then
7:        $\mathbf{Q}_g \leftarrow \text{Breed}(\text{parents})$ 
8:     end if
9:     if morphic enabled and  $\mathcal{F}.\text{count} > 0$  then
10:       $\mathbf{Q}_f \leftarrow \mathcal{F}.\text{emit}(s = 0.5)$ 
11:    end if
12:     $\mathbf{Q}_{\text{init}} \leftarrow \text{Blend}(\mathbf{Q}_g, \mathbf{Q}_f)$   $\triangleright 50/50$  if both, else whichever is non-null, else zeros
13:     $a_i \leftarrow \text{Agent}(M, \mathbf{Q}_{\text{init}})$ 
14:     $a_i.\text{train}(E)$ 
15:  end for
16:  evals  $\leftarrow [a_i.\text{evaluate}() \text{ for } i = 1 \dots P]$ 
17:   $h_g \leftarrow \text{mean}(\text{evals})$ 
18:  if morphic enabled then
19:    for each  $(a_i, e_i) \in (\text{agents}, \text{evals})$  do
20:       $\mathcal{F}.\text{absorb}(a_i.\mathbf{Q}, w = \text{perf\_weight}(e_i))$ 
21:    end for
22:  end if
23:  if genetic enabled then
24:    parents  $\leftarrow$  top 30% of agents by eval performance
25:  end if
26: end for
27: return  $H$ 
```

Control. Remains flat at 400 steps across all 60 generations. Individual Q-learning agents cannot solve this maze within the 100-episode training budget starting from a zero Q-table. This establishes a firm baseline.

Genetic Only. Converges rapidly to approximately 56 steps by generation 5 and remains stable. Elite selection with Q-table crossover is a powerful optimization mechanism, consistent with evolutionary computation results.

Morphic Field Only. This is the key morphic resonance result. Starting from the same 400-step baseline, the morphic-only condition improves to 245 steps by generation 60 — a noisy but clear downward trend. Learning occurs across generations with **no genetic inheritance**. The only channel is the shared morphic field.

The improvement is slower and noisier than the genetic condition. This is expected: the morphic field is a blunt instrument compared to elite selection. It averages over all past agents (performance-weighted), while genetic inheritance directly copies successful Q-tables with targeted mutation.

Genetic + Morphic. Intermediate at 108 steps — better than morphic-only, worse than genetic-only. The 50/50 blending dilutes the genetic signal with the (noisier) morphic prior in early generations when the field is immature.

4.2 The Displaced Control

The displaced control is the most important result. Fresh agents with **zero genetic connection** to any prior solver are initialized using only the mature morphic field (accumulated over 60 generations of morphic-only training with a separate seed).

- **Generation 1:** 297 steps (vs. 400-step control baseline).
- **Generation 20:** 228 steps.
- **Immediate advantage:** 103 steps (26%) on the very first generation, with zero ancestry.

These agents benefit purely because *other agents solved this maze before*. There is no genetic inheritance, no training history, no shared environment — only the accumulated morphic field.

This directly models Sheldrake’s interpretation of the Agar-Crew data: unrelated rat lines (control groups never selected for maze ability) also improved over time, suggesting a non-genetic channel of information transfer.

4.3 Temporal Decay Comparison

We compare the morphic field with five decay rates ($\rho \in \{0, 0.005, 0.01, 0.02, 0.05\}$) to determine whether temporal decay improves performance. Moderate decay ($\rho = 0.01$ – 0.02) prevents stale early-generation knowledge from diluting recent improvements, while high decay ($\rho = 0.05$) forgets too aggressively. The optimal decay rate balances recency with cumulative depth — precisely the trade-off that pheromone systems solve in biological stigmergy.

4.4 Phase Transition: Population Threshold

We vary population size ($P \in \{1, 2, 5, 10, 20, 40\}$) while keeping all other parameters constant to test whether the morphic field exhibits a phase transition — a critical population threshold below which the field fails to form coherent knowledge. This experiment connects to Khushiyant’s (2025) finding of phase transitions in stigmergic collective memory at critical density $\rho \approx 0.20$ – 0.23 [27].

4.5 Adversarial Robustness

We test the morphic field’s resilience to adversarial Q-value injection by comparing an undefended **MorphicField** against a **RobustMorphicField** with consensus filtering. Three attack types are tested at the midpoint of training: (1) random noise injection with high performance weight, (2) degenerate policy injection (always-left Q-tables), and (3) zero-flood attacks that dilute the field. The consensus filter rejects Q-tables whose mean deviation from the running field statistics exceeds a threshold, preventing adversarial contributions from corrupting the accumulated knowledge.

4.6 Multi-Maze Generalization

We test whether the morphic field captures general maze-solving strategies or merely memorizes Q-values for a specific topology. Two experiments: (1) *Same-size transfer*: build the field on Maze A (seed=42) and test displaced agents on Mazes B, C, D (different seeds, same size); (2) *Cross-maze field*: train across four mazes simultaneously, alternating each generation, then test on all four plus a held-out maze.

4.7 Deep RL Extension

To demonstrate that the morphic field pattern scales beyond tabular Q-learning, we implement a numpy-only DQN (2-layer neural network, 64 hidden units, experience replay) and test the morphic field as shared network weight initialization. The `DeepMorphicField` accumulates and emits network parameters ($\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2$) using the same performance-weighted absorb/emit cycle.

5 Epistemological Analysis

5.1 What the Simulation Proves

The simulation demonstrates three things:

1. **Formal coherence.** The hypothesis of morphic resonance can be formalized as a mathematical model — specifically, a performance-weighted shared Q-table with absorb/emit operations — and this model produces learning across generations with no genetic channel. The hypothesis is not incoherent or self-contradictory.
2. **Distinctive data pattern.** If morphic resonance were real, the data pattern would be distinctive: unrelated agents (or organisms) solving a task faster purely because other agents solved it before, with no physical channel of inheritance. This pattern is detectable and distinguishable from genetic selection (which requires ancestry) and individual learning (which cannot exceed the single-generation control baseline).
3. **Experimental design validation.** The displaced control test is the most powerful experimental design for detecting morphic resonance. If fresh, unrelated agents outperform the control baseline on their first generation, something other than genetics or individual learning must be transmitting information.

5.2 What the Simulation Does Not Prove

The simulation does **not** demonstrate:

1. **That morphic resonance exists in nature.** In the simulation, we *built* the field — it is a real data structure (`MorphicField` class). We know exactly how information gets from past agents to present ones, because we wrote the code. In Sheldrake’s hypothesis, the field’s physical existence and mechanism are unknown.
2. **That information can transmit without a physical channel.** The morphic field in the simulation *is* a physical channel — it is a NumPy array in computer memory. The simulation shows what would happen *if* such a channel existed, not that one does.
3. **That Sheldrake’s specific mechanism is real.** The simulation implements *a* formalization of morphic resonance, not *the* formalization. Sheldrake’s description is qualitative; many formalizations are possible.

The analogy. The epistemic status is: “If gravity worked like an inverse-cube law, orbits would look like *X*” (a simulation) versus “Gravity actually is inverse-cube” (an empirical claim). We have demonstrated the *if*, not the *is*.

5.3 The Critical Epistemic Difference

In the simulation, the morphic field is a *design choice* — a data structure we implemented. In nature, whether any such field exists is an *empirical question*. The simulation makes the hypothesis precise enough to test: run a McDougall-style experiment with rigorous displaced

controls; if the predicted pattern appears (unrelated lines improving), the hypothesis gains evidence. If it does not, the hypothesis is weakened. Either way, the simulation has served its purpose by sharpening the prediction.

6 Applications to Multi-Agent AI Systems

Independent of whether morphic resonance exists in nature, the morphic field architecture — performance-weighted absorption, scaled emission, blending with local learning — constitutes a practical design pattern for AI systems. This section strips away the Sheldrake framing and examines the engineering utility.

6.1 Population-Level Experience Replay

The morphic field is a population-level experience replay buffer where contributions are weighted by success. Unlike standard experience replay [12], which stores individual transitions for a single agent, the morphic field operates at the *policy level*: it accumulates entire Q-tables, weighted by evaluation performance, and emits a distilled initialization for new agents.

This is related to but distinct from:

- **Knowledge distillation** [13]: compresses many models into one. The morphic field does not compress — it accumulates and averages.
- **Federated learning** [15]: aggregates local updates into a global model. The morphic field is structurally similar, but asymmetric — performance weighting means not all agents contribute equally.
- **Population-based training** [14]: evolves hyperparameters across a population. The morphic field evolves policies, not hyperparameters, and includes non-elite contributors (at reduced weight).

6.2 Cold-Start Mitigation

The displaced control result — fresh agents immediately performing 26% better — is a direct solution to the *cold-start problem*. A new agent spins up with no history but benefits from a collective memory accumulated by all prior agents.

Applications include:

- **Scaling agent swarms**: new workers bootstrap from the swarm’s accumulated knowledge rather than learning from scratch.
- **Recovery after failure**: a crashed agent’s replacement inherits the field immediately.
- **Cross-task transfer**: the field serves as a domain prior (rather than task-specific weights) that gives new agents a head start on related problems.

6.3 Stigmergic Coordination

The morphic field is a form of *stigmergy* [16] — agents coordinate through a shared environment rather than direct communication. This is the same principle by which ant colonies optimize paths [17], Wikipedia edits accumulate knowledge, and git repositories coordinate distributed development.

Stigmergic architectures scale better than peer-to-peer messaging because: (A) agents do not need to know about each other; (B) the shared medium naturally aggregates and attenuates information; (C) new agents benefit immediately without explicit onboarding.

Connection to GraphPalace. The GraphPalace stigmergic memory engine [18] uses five-type pheromone trails for memory retrieval: success, traversal, recency, exploitation, and exploration pheromones accumulate on graph edges to create living retrieval landscapes. The morphic field implements the same principle in Q-table space: agents deposit performance-weighted Q-values and withdraw scaled priors. A natural extension would apply morphic field architecture to GraphPalace’s embedding space, creating a “learning palace” where the memory system itself improves from the agents that use it.

6.4 Summary: The Morphic Field as a Design Pattern

Table 3 summarizes the morphic field as a reusable architecture.

Table 3: The morphic field as a design pattern for multi-agent systems.

Component	Description
Shared medium	Accumulated representation (Q-table, embedding, weight matrix)
Absorb	Performance-weighted deposit after each agent’s evaluation
Emit	Scaled prior for new agents (strength parameter controls influence)
Blend	Combination with local/genetic initialization (tunable ratio)
Performance gate	Failed agents contribute near-zero; solvers contribute proportionally
No identity	The field does not track individual contributors — only the aggregate

The pattern is agnostic to the representation space. The simulation uses flat Q-tables; the same absorb/emit/blend cycle could operate on:

- Neural network weight matrices (deep RL agents)
- Embedding vectors (semantic memory systems like GraphPalace)
- Tool-use strategy distributions (LLM agent systems)
- Hierarchical fields (different fields for different skill levels, mapping to GraphPalace’s wing/-room/drawer hierarchy)

7 Discussion

7.1 Limitations

1. **Scale.** While the deep RL extension (Section 4.7) demonstrates the pattern scales beyond tabular Q-learning, it uses a small numpy-only DQN. Production validation with PyTorch/TensorFlow on continuous state spaces remains future work.
2. **Morphic-only noise.** The morphic-only condition is noisy (39% improvement vs. 86% for genetic). Performance-weighted averaging is a blunt aggregation mechanism compared to elite selection with crossover.
3. **No real biology.** The simulation models artificial agents, not rats. The mapping from Q-learning to animal learning is loose at best, though Levin’s bioelectric field work [24] provides a biological bridge.
4. **Sheldrake’s mechanism is unspecified.** Our formalization (shared Q-table) is one of many possible implementations. Sheldrake does not specify the mathematical structure of morphic fields, so our results are specific to this formalization.

5. **Single seed.** Primary results are from a single random seed. The multi-maze experiments (Section 4.6) test across topologies, but a full study would require multiple seeds with bootstrap confidence intervals.

7.2 Criticisms of Morphic Resonance

For completeness, we note the major scientific criticisms of Sheldrake’s hypothesis, which our simulation does not address:

1. **No known mechanism:** the hypothesis posits influence across time and space without an energetic carrier or physical mechanism [19].
2. **Violates established physics:** implies non-local causation without known fields, contradicting the standard model.
3. **Unfalsifiable boundaries:** critics argue the hypothesis can always be adjusted to accommodate negative results [4].
4. **Alternative explanations:** genetic drift, methodological artifacts, and publication bias can explain the cited experimental results without invoking new physics.

7.3 Future Work

1. **Production-scale deep RL:** extend the numpy DQN proof of concept to PyTorch/TensorFlow on continuous state spaces (Atari, MuJoCo) with proper target networks and prioritized replay.
2. **Embedding-space fields:** implement the absorb/emit/blend pattern in vector embedding space (e.g., within GraphPalace) to create stigmergic memory systems that improve from agent usage.
3. **Multi-seed statistical analysis:** run all conditions across 50+ seeds with bootstrap confidence intervals.
4. **Head-to-head FedRL comparison:** benchmark morphic field directly against FedAvg [15], FEDORA [23], and SUPER [21] on identical cooperative tasks with identical compute budgets.
5. **LLM agent fields:** apply the morphic field pattern to LLM tool-use strategy distributions, where agents share accumulated knowledge about which tools to invoke for which tasks.

8 Conclusion

We have presented a computational model of Sheldrake’s morphic resonance hypothesis, implemented as a multi-agent Q-learning simulation of the McDougall rat maze experiment, and extended it across five additional experimental axes. The key findings:

1. A morphic field, formalized as a performance-weighted shared Q-table, transmits useful learning across agent generations with no genetic channel (39% improvement over 60 generations in the morphic-only condition).
2. The displaced control test produces a distinctive signature: fresh agents with zero ancestry immediately outperform the control baseline by 26%, benefiting purely from the accumulated collective habit.
3. Temporal decay ($\rho = 0.01\text{--}0.02$) improves field quality by preventing stale knowledge from dominating — a mechanism absent from all existing FedRL methods.

4. The morphic field exhibits population-dependent effectiveness, with a clear threshold below which the field fails to form coherent knowledge.
5. Consensus filtering provides robustness against adversarial Q-value injection, with up to 14% recovery against zero-flood attacks.
6. The morphic field architecture extends to neural Q-networks and transfers partially across maze topologies, demonstrating generality beyond the tabular setting.

The hypothesis is formally coherent, the predicted data pattern is distinctive and detectable, and the displaced control constitutes a viable experimental design. The simulation does not prove that morphic resonance exists in nature — it is a computational thought experiment that makes the hypothesis precise enough to test empirically.

The engineering contribution stands independently. The morphic field pattern — value-function-level sharing with performance weighting, temporal decay, and consensus filtering — occupies a specific gap between experience-sharing (SUPER) and weight-sharing (FedAvg) in the multi-agent knowledge transfer spectrum. The insight that learning should not die with the agent — that collective memory can be infrastructure — provides a clean, minimal implementation with a direct path from biological inspiration to practical multi-agent architecture.

Code availability. The simulation code, experiment scripts, and paper source are available at <https://github.com/web3guru888/morphic-resonance-sim>. Dependencies: Python 3, NumPy, Matplotlib.

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